

Yuxing Liang

Ph.D. Candidate in Economics | Concordia University

yuxing.liang@mail.concordia.ca | yuxing-liang.github.io | +1 514 431 2815 | Montréal, Canada

Education

Ph.D. in Economics, Concordia University, Montréal, Canada

Expected December 2026

Supervisor: Szilvia Pápai

External research advisor: Xintong Han

M.A. in Economics, Concordia University, Montréal, Canada

2018–2020

B.A. in Economics, Shanghai Customs College, Shanghai, China

2014–2018

Research Profile

Fields: Market Design, Matching Theory, Applied Microeconomic Theory

Research interests: Strategy-proofness, preference elicitation, assignment mechanisms, school choice, college and major assignment

Methods: Computational analysis, simulation, empirical analysis

Job Market Paper

Why Students Can Misreport under Strategy-Proof Matching

Strategy-proofness makes truth-telling optimal, yet students often submit nontruthful rankings. This paper asks which comparisons in a ranking assignment incentives actually identify. The key intuition is menu exposure: a comparison is identified only when rivals can generate a menu on which that comparison changes the student's assignment. This insight yields a new sharp capacity boundary for mandatory complete rankings. Under Deferred Acceptance and Top Trading Cycles, if total capacity exceeds enrollment—even by one seat—the dominant reports are exactly those truthful through a student-specific guarantee rank; the same boundary extends to a broader class of strategy-proof mechanisms. The spare seat allows every rival to be assigned while two schools remain available, making each comparison above the guarantee potentially decisive. Without slack, this exposure can fail and dominant reports need not form a truthful prefix. The boundary disappears when truncation allows rivals to exit. Reanalyzing public Chen–Sönmez DA data, I separate 58 nontruthful rankings into six dominance-safe deviations, 32 that are harmless only at the realized profile, and 20 that change the assignment.

Working Papers

Towards a Better Major Assignment: An Empirical Study with Students in China (with Xintong Han and Yan Zeng), submitted

Using students' submitted rankings, exam records, and major characteristics from a Chinese university, we estimate preferences and compare assignment reforms through counterfactual analysis. The results show how alternative mechanisms affect welfare, first-choice placement, and the distributional consequences of preferential treatment.

Progressive Choice Mechanisms in School Choice (with Szilvia Pápai)

We study a family of mechanisms that varies how many schools each student may consider at each step, connecting Immediate Acceptance, Chinese parallel mechanisms, Serial Dictatorship, and Deferred Acceptance. We establish efficiency under common priorities, show that broader choice opportunities weakly reduce manipulability, and study the corresponding stability trade-offs.

Work in Progress

Bridging Theory and Evidence: How External Information Transforms Students' Application Decisions

This project studies how feedback about relative rank changes students' stated preferences and applications. It connects information acquisition to assignment design by tracing how students revise the comparisons that shape their submitted rankings.

Selected Presentations

<i>Why Students Can Misreport under Strategy-Proof Matching</i> . 18th Meeting of the Society for Social Choice and Welfare, University of Tokyo	June 2026
<i>Why Students Can Misreport under Strategy-Proof Matching</i> . 20th CIREQ–CIRANO Ph.D. Students' Conference, CIRANO	May 2026
<i>Why Students Can Misreport under Strategy-Proof Matching</i> . CIREQ Interdisciplinary Ph.D. Conference in Applied Market Design, Concordia University	March 2026
<i>Progressive Choice Mechanisms in School Choice</i> . Management Science Workshop Series (virtual), Universidad Católica de Chile	October 2025
<i>Progressive Choice Mechanisms in School Choice</i> . 14th Conference on Economic Design, University of Essex	June 2025
<i>Progressive Choice Mechanisms in School Choice</i> . 17th Meeting of the Society for Social Choice and Welfare, Paris School of Economics	July 2024
<i>Towards a Better Major Assignment: An Empirical Study with Students in China</i> . Economics Department Lunch Seminar, Concordia University	November 2023
<i>Towards a Better Major Assignment: An Empirical Study with Students in China</i> . Urban and Education Economics Seminars, Shandong University	July 2023
<i>Towards a Better Major Assignment: An Empirical Study with Students in China</i> . 13th Conference on Economic Design, Universitat de Girona	June 2023
<i>Towards a Better Major Assignment: An Empirical Study with Students in China</i> . 18th CIREQ Ph.D. Students' Conference, Concordia University	May 2023
<i>Progressive Choice Mechanisms in School Choice</i> . Agents' Behavior in Combinatorial Game Theory, CRM Montréal (virtual)	November 2021

Course Instructor, Concordia University

- **ECON 332** Introductory Financial Economics, Winter 2027 (scheduled)
- **ECON 436/536** Economics of Taxation, Fall 2026
- **ECON 491/591** Environmental Economics, Winter 2026
- **ECON 221** Statistical Methods I, Winter 2025
- **ECON 377** The Asia-Pacific Rim Economies, Winter 2024

Tutorials and Teaching Assistantships, Concordia University

Tutorial Leader

- **ECON 612** Microeconomics I, Winter 2023

Teaching Assistant

Graduate courses

- **ECON 614** Game Theory, Winter 2022 and Fall 2025
- **ECON 642** Financial Economics I, Fall 2024
- **ECON 695** Empirical Industrial Organization, Fall 2023
- **ECON 612** Microeconomics I, Winter 2023
- **ECON 673** Industrial Organization, Fall 2022

Undergraduate courses

- **ECON 211** Statistical Methods I, Fall 2025
- **ECON 301** Intermediate Microeconomic Theory I, Fall 2023
- **ECON 485** Health Economics, Winter and Fall 2023
- **ECON 461/561** Industrial Organization, Winter 2023
- **ECON 464** Game Theory, Fall 2022
- **ECON 422** Econometrics II, Winter 2022
- **ECON 203** Macroeconomics, 2020–2025
- **ECON 201** Microeconomics, 2020–2025

Research Experience

Research Assistant, Concordia University

Winter 2023

Prepared research datasets, conducted literature reviews, and supported statistical analysis for faculty research.

Research Assistant, Shanghai Customs College

2017–2018

Collected and cleaned research data, reviewed related literature, and supported descriptive empirical analysis.

Computational Methods for Matching Research

Mechanism implementation and simulation

MATLAB, Python

Implement and compare Serial Dictatorship, Immediate Acceptance, Deferred Acceptance, Top Trading Cycles, and Progressive Choice mechanisms; simulate finite markets and diagnose allocation outcomes.

Preference and assignment data analysis

Python, Stata, R

Construct reproducible rank-order-list and assignment datasets, estimate discrete-choice models, and evaluate mechanism-based counterfactuals.

Computational support for theory

Python, MATLAB

Enumerate finite markets, search for counterexamples, stress-test conjectures, and verify theoretical boundaries across market environments.

Selected Grants and Awards

FRQSC Doctoral Research Scholarship (B2Z), Fonds de recherche du Québec	CAD \$58,333	2023–2026
Graduate Fellowship, Concordia University	CAD \$56,000	2020–2024
International Tuition Award of Excellence, Concordia University	CAD \$40,656	2020–2024
Professor Jaleel Ahmad Scholarship, Concordia University	CAD \$2,500	2021–2022
SGS Conference and Exposition Allowance, Concordia University	CAD \$2,000	2023, 2024
Research Bursary, CIREQ	CAD \$500	2023
Conference Travel Funding, Faculty of Arts and Science, Concordia University	CAD \$400	2023
Conference Travel Funding, Department of Economics, Concordia University	CAD \$400	2023, 2024
Outstanding Undergraduate Thesis, Shanghai Customs College		2018

Additional Professional Experience

Analyst, KYC Periodic Team, KPMG Advisory (China) Limited, Shanghai

Jan. 2017–Feb. 2018

Market Research Assistant, Tebon Securities Co., Ltd., Shanghai

Aug. 2016–Jan. 2017

Technical Skills and Languages

Programming and analysis: Python, MATLAB, Stata, R, EViews, SAS; PostgreSQL, MySQL

Research workflow: L^AT_EX, GitHub, Microsoft Office, Power BI, Tableau

Languages: Mandarin (native), English (fluent), Japanese (intermediate), French (beginner)

References

Szilvia Pápai

Professor
Department of Economics
Concordia University
szilvia.papai@concordia.ca

Ming Li

Professor
Department of Economics
Concordia University
ming.li@concordia.ca

Huan Xie

Professor
Department of Economics
Concordia University
huan.xie@concordia.ca